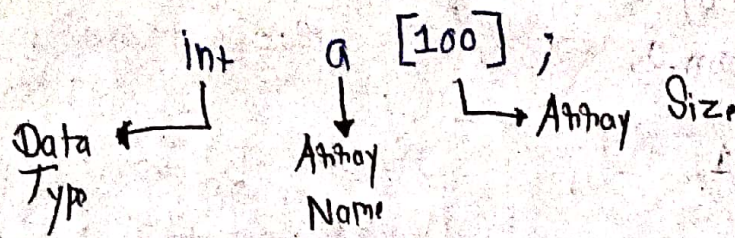
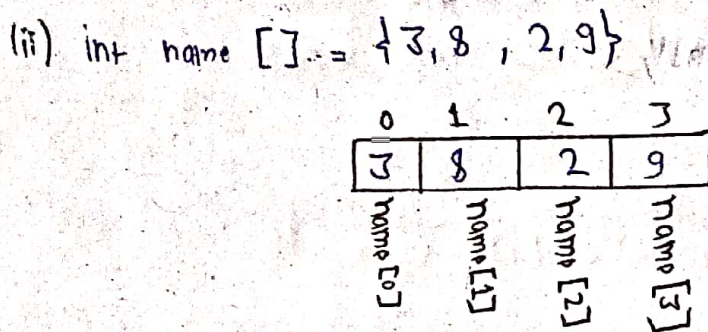
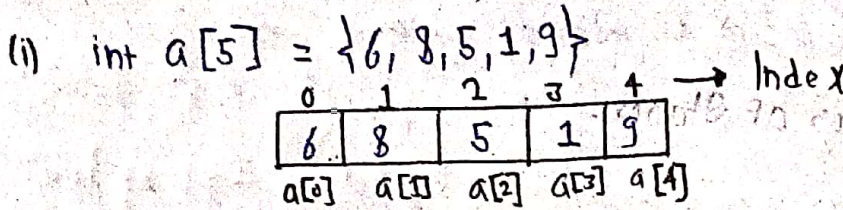


• ARRAY:-

An array is a collection of elements of same data type stored in contiguous memory locations and accessed using an index.

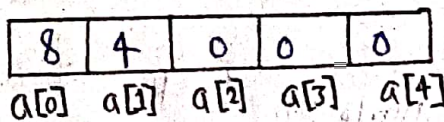


Ex →



(iii) `int arr[10];`

(iv) `int a[5] = {8, 4}`



Q) Find the index of a specific element in an array, if the element is not present, print -1. Ask the size of the array from the user.

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
int main () {
```

```
    int n;
```

```
    cout << "Enter the length of the array:";
```

```

cin >> n;
int arr[n];
cout << "Enter" << n << " Elements: ";
for(int i=0; i<n; i++) {
    cin >> arr[i];
}

```

```

int num;
cout << "Enter the element to Search: ";
cin >> num;

for(int i=0; i<n; i++) {
    if(arr[i] == num) {
        cout << "Element " << num << " Found! at index " << i << endl;
        return 0;
    }
}

```

```

cout << "Element Not Found!" << endl;
return 0;
}

```

Q) Find the Second largest element in an array of Unique elements of size n, where $n > 3$.

```

#include <bits/stdc++.h>
using namespace std;

```

```

int main() {
    int n;
    cout << "Enter the length of the array: ";
    cin >> n;
    int arr[n];
    cout << "Enter" << n << " Elements: ";
    for(int i=0; i<n; i++) {
        cin >> arr[i];
    }
}

```

int largest, Second-largest;

if (arr[0] > arr[1]) {

largest = arr[0];

Second-largest = arr[1];

else {

largest = arr[1];

Second-largest = arr[0];

// Comparing with all remaining elements

for (int i = 2; i < n; i++) {

if (arr[i] > largest) {

Second-largest = largest;

largest = arr[i];

else if (arr[i] > Second-largest) {

Second-largest = arr[i];

}

cout << "Largest: " << largest << "\n Second Largest: " << Second-largest << endl;

return 0;

}

Q) Find the third Smallest element in an array of Unique elements Size n. where $n > 3$

#include <bits/stdc++.h>

using namespace std;

int main () {

int n;

cout << "Enter the length of array: ";

cin >> n;

```

int arr[n];
cout << "Enter" << n << " Elements: ";
for(int i=0; i<n; i++) {
    cin >> arr[i];
}

int Smallest = INT_MAX;
int Second-Smallest = INT_MAX;
int Third-Smallest = INT_MAX;

for(int i=0; i<n; i++) {
    if (arr[i] < Smallest) {
        Third-Smallest = Second-Smallest;
        Second-Smallest = Smallest;
        Smallest = arr[i];
    }
    else if (arr[i] < Second-Smallest) {
        Third-Smallest = Second-Smallest;
        Second-Smallest = arr[i];
    }
    else if (arr[i] < Third-Smallest) {
        Third-Smallest = arr[i];
    }
}

cout << "Third Smallest: " << Third-Smallest << endl;
return 0;
}

```

Q> WAP to reverse an array

```

#include <bits/stdc++.h>
using namespace std;

```

```

int main() {

```

```

    int n;

```

```

    cout << "Enter the length of the array: ";

```

```

    cin >> n;

```

```

    int arr1[n], arr2[n];

```